Leveling up
In Coding Speed

Arman Ferdous

https://codeforces.com/problemset/problem/2188/B

B. Seats

time limit per test: 1 second
memory limit per test: 256 megabytes

Cordell manages a row of n seats at the *Scuola Comunale di Musica Piova* where students are strictly forbidden from sitting next to each other.

You are given a binary string* s, where $s_i = 1$ indicates that the i-th seat has been occupied by a student, and $s_i = 0$ indicates that it is free now. It is guaranteed that no two adjacent seats are occupied currently. Cordell needs to add more students until it is impossible to seat anyone else in the row. However, she wants to achieve this state with as few students as possible.

Your task is to calculate the minimum **total** number of students seated when it is impossible to seat anyone else in the row.

*A binary string is a string where each character is either **0** or **1**.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the number of seats in the row.

The second line of each test case contains the binary string s of length n ($s_i \in \{0, 1\}$). It is guaranteed that no two adjacent characters are both 1.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the minimum total number of seated students.

https://codeforces.com/problemset/problem/2176/B

B. Optimal Shifts

time limit per test: 1.5 seconds
memory limit per test: 256 megabytes

You are given a binary string $s_1 s_2 \ldots s_n$, containing at least one `1`. You want to obtain a binary string of the same length, consisting only of `1`s. To do this, you can perform the following operation any number of times:

Choose a number d ($1 \le d \le n$) and consider the string t as a cyclic right shift of the string s by d, or, more formally, $t = s_{n-d+1} \ldots s_n s_1 \ldots s_{n-d}$. After that, for all j for which $t_j = 1$, perform $s_j := 1$. The described operation costs d coins, where d is the chosen shift amount.

Note that the positions j in the string s, where initially $s_j = 1$, remain equal to 1 even if $t_j = 0$.

You need to determine the minimum number of coins that can be spent so that the string s consists only of `1`s after all operations.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains the number n ($1 \le n \le 2 \cdot 10^5$) — the size of the given binary string.

The second line of each test case contains a binary string of length n, each element of which is either `0` or `1`.

It is guaranteed that at least one character in each string is equal to `1`.

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the answer to it — the minimum possible number of coins that you can spend to make all characters in the string equal to `1`.

https://codeforces.com/contest/2245/problem/B

B. Delete and Concatenate

time limit per test: 1.5 seconds
memory limit per test: 256 megabytes

You are given an array a of n integers. Your initial score is 0.

You can perform the following two types of operations any number of times **until the array becomes empty**:

- Choose a single element. Add its value to your score, and remove it from the array.
- Choose two adjacent elements. Add the maximum of their values to your score, and remove both elements from the array. (This operation requires the array to have at least 2 elements.)

After each operation, the remaining elements are concatenated without changing their relative order.

You are also given an integer c, which represents the cost of performing one operation. Every time you perform an operation (regardless of the type), c is subtracted from your score.

Compute the maximum possible final score you can achieve.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and c ($1 \le n \le 2 \cdot 10^5$, $0 \le c \le 10^9$), representing the length of a and the cost of performing one operation, respectively.

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($-10^9 \le a_i \le 10^9$), representing the elements of a.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

https://codeforces.com/problemset/problem/1913/B

B. Swap and Delete

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a binary string s (a string consisting only of `0`-s and `1`-s).

You can perform two types of operations on s:

1. delete one character from s. This operation costs 1 coin;
2. swap any pair of characters in s. This operation is free (costs 0 coins).

You can perform these operations any number of times and in any order.

Let's name a string you've got after performing operations above as t. The string t is *good* if for each i from 1 **to** $|t|$ $t_i \neq s_i$ ($|t|$ is the length of the string t). The empty string is *always good*. Note that you are comparing the resulting string t with the *initial string* s.

What is the minimum total cost to make the string t good?

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases. Then t test cases follow.

The only line of each test case contains a binary string s ($1 \leq |s| \leq 2 \cdot 10^5$; $s_i \in \{$`0`, `1`$\}$) — the initial string, consisting of characters `0` and/or `1`.

Additional constraint on the input: the total length of all strings s doesn't exceed $2 \cdot 10^5$.

Output

For each test case, print one integer — the minimum total cost to make string t good.

https://codeforces.com/problemset/problem/1883/C

C. Raspberries

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given an array of integers $a_1, a_2, \ldots, a_n$ and a number k ($2 \leq k \leq 5$). In one operation, you can do the following:

- Choose an index $1 \leq i \leq n$,
- Set $a_i = a_i + 1$.

Find the minimum number of operations needed to make the product of all the numbers in the array $a_1 \cdot a_2 \cdot \ldots \cdot a_n$ divisible by k.

Input

Each test consists of multiple test cases. The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases. Then follows the description of the test cases.

The first line of each test case contains two integers n and k ($2 \leq n \leq 10^5$, $2 \leq k \leq 5$) — the size of the array a and the number k.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq 10$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the minimum number of operations needed to make the product of all the numbers in the array divisible by k.

https://codeforces.com/contest/2233/problem/C

C. Cost of a Bracket Sequence

time limit per test: 2 seconds
memory limit per test: 512 megabytes

Let the cost of an arbitrary bracket string be the length of it's longest subsequence* that is a regular bracket sequence†.

You are given a bracket string s and an integer k. Your task is to remove at most k characters from the string s so that the cost of the resulting string is minimized.

---

*A sequence a is a subsequence of a sequence b if a can be obtained from b by the deletion of several (possibly, zero or all) element from arbitrary positions.

†A bracket sequence is called regular if it is possible to obtain a correct arithmetic expression by inserting the characters $+$ and 1 into this sequence. For example, the sequences "(())()", "()", and "((()))" are regular, while ")(", "(()", and "(())(" — are not.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^3$). The description of the test cases follows.

The first line of each test case contains two integers n and k ($1 \le n \le 5\,000$; $0 \le k \le n$) — the length of the string s and the maximum number of deletions.

The second line of each test case contains a string s of length n consisting of the characters "(" and/or ")".

Additional input constraints:

- the sum of n over all test cases does not exceed $5\,000$.

Some Classical Techniques

- Prefix Sums
- Difference Arrays
- Previous Greater Element

Maximum Subarray Sum

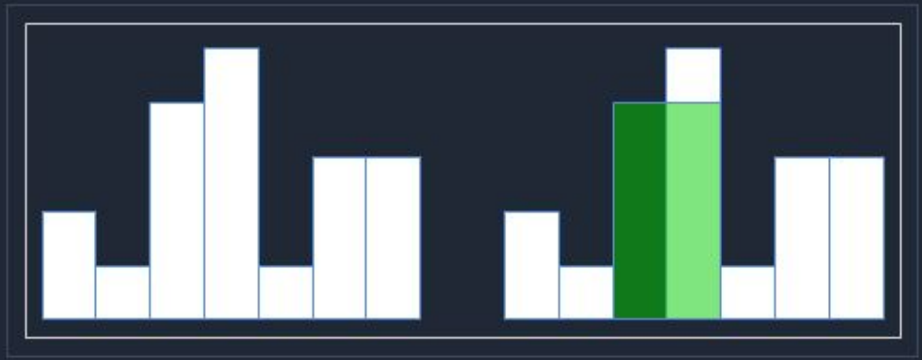
- Given an array of integers, output the maximum possible sum of any subarray.

Contribution Method

- Given an array of integers, evaluate the sum of min(A[L], …, A[R]) for all pairs L <= R.

Largest Rectangle in Histogram - LightOJ 1083

A histogram is a polygon composed of a sequence of rectangles aligned at a common base line. The rectangles have equal widths but may have different heights. For example, the figure shows the histogram that consists of rectangles with the heights `{2, 1, 4, 5, 1, 3, 3}` measured in units where the width of the rectangles is 1.



Usually, histograms are used to represent discrete distributions, e.g., the frequencies of characters in texts. Note that the order of the rectangles, i.e., their heights, is important. Calculate the area of the largest rectangle in a histogram that is aligned at the common base line, too. The figure on the right shows the largest aligned rectangle for the depicted histogram.

Largest sub-rectangle of 1s.

- Given a binary matrix, find the largest sub-rectangle in it which only contains 1s.

This week's homework

- Finish LightOJ Beginner section (all the important problems)
- Participate in at least 2 contests (cf, cc, ac) and solve at least 2 probs in each contest.